

Index of Coincidence for Each Substring

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This document provides the results of the Index of Coincidence (IC) analysis for the given cipher text with guessed keyword lengths of 6, 7, and 8. The IC values help in estimating the likelihood of a guessed keyword length being correct by comparing the structure of substrings with the expected distribution of letters in English.

The Index of Coincidence (IC) values calculated for each substring with guessed keyword lengths are as follows:

- For $m = 6$, the IC values are approximately 0.04228753702437913, 0.042606516290726815, 0.0405559352927774, 0.0405559352927774, 0.04205969469127364, 0.04201412622465254

- For $m = 7$, the IC values are approximately 0.071260086902545, 0.071260086902545, 0.06815642458100558, 0.061390440720049656, 0.06834264432029795, 0.06225946617008069, 0.06790813159528243

- For $m = 8$, the IC values are approximately 0.04031282754172378, 0.03982907361122309, 0.043699105055228574, 0.04329597677981133, 0.04172791115466275, 0.044177690674505964, 0.04050302139474114, 0.037889923240241714

An IC value close to 0.065 suggests that the substring is less random and more structured, resembling the distribution of a standard text in the corresponding language. This indicates that the substrings encrypted with the same key letter (if the guessed keyword length is correct) reveal a structure that resembles the natural frequency of letters in English.

Based on the calculated IC values, the analysis supports our guess that $m = 7$ is the correct keyword length for the ciphered text. This is because the IC values for $m = 7$ are the highest and most consistent, indicating a significant similarity to the expected distribution of letters in English text, and thus revealing a structure indicative of correct keyword length.

Substring y1:

LJXWFDTLJFTVLKWFEGWSDSSUWVVLWKLJLVJSUKEDGUGJELGUQWFKMJRSRWVPSM
GAWUUUTDZULHELDZGVAWZSENWLWGLWUAEKGKJWGMGJUJZWLDZUWWLJFSWLW
KLWNZDFLKTUZYGYLLWKVJJVLSSWLAHGLWWXVMSWOVYSJDLZLULUFJVZ

Index of Coincidence: 0.0713

Substring y2:

AMCATTXMHZLRLKHPAGXBFMKXBALHKTHTXPXLXXOHYHFBFAIKKXKHNGWXZTGGG
VGXHTRMXXHMFBBXXYNLLXMHXKMKHVWYFBOLBKDLGGHTHXVAZMHMTTWAMHGKLWBA
QXTEZAGXBLBGPALMBHHHGVWLAIXGAUYLXMEMBHAGFBXHHMRNUEHG

Index of Coincidence: 0.0713

Substring y3:

VDLRECTVKKIVFNVEVRVJEGJMJERLCFTKKJVDFJKVEVELFKEIVVKKIKUKSIVDPZJVTIDKU
NBJUKLFCYKJRSEKIFNREKVKLFZVKRZVZJMEMURREVXYETSZPKPSRZFZVSIFNVFVJKVJ
NFTJIDMLKTRYURVCKTRVFFSNICYCEFUVTNRNGCRIFRLF

Index of Coincidence: 0.0682

Substring y4:

HIWWHASHILXGJLVGXZZAIWVEMMEXWIPGLEYSVXPMRRLSSZWSSHLAMCEJTGGW
WVXQMIMIWIWLRCELITSXMMIRSXEVLRRHVLRRGGXMRMEYWIGCERGCVALWYRWRRGVI
RLVXRMLPIVSMCMETLVTXVPQIMIXIGQEIEEPKRMHHFFVMRMXAVP

Index of Coincidence: 0.0614

Substring y5:

ENTBWAMTCEYUTOVEHEESETONCGWGECOEELFNENRTEECISELVEPARUOASALOLOAE
NYAUOFTYACEISINERPTH SOLIRRNOIEISEVATGOAHDODTSDDHSTOEADAILDCSTUASGI
EYAOEKAMKNNACCSEAPHELUCNRHNUUTCGTNOENPEOEODTHSTO

Index of Coincidence: 0.0683

Substring y6:

BFUQUKYAFXMDTYQZEDAOZDPPQZMUDAHEUEFAFFMQTZOOORFAUZAOPXDDERVDQYF
DABUZZMTZDDOOEZPQFXTQUZQOWQQZOOOBDQIEFYFQQFQMQQFZGUFEZBPHXXFGIT
FZFAXTQPZDXFGISPOEKMDRDMEAXASZMFEOQMDIQUERGNGEFQEKAMQS

Index of Coincidence: 0.0623

Substring y7:

SGUWDKAHAFAFGOALZOFSEWMKKWQNKFWKJSDJAATUWLLGSJWYVAFGWQAJMWMWE
EANOLFAKMLWQQGSGLKFZAOVFGUFKKDAWGSJAVSWZEAHJHLTAKZGULTKQSGWNWZK
AWAUAAWSLWOLWAKAJMSHYFKLGLKXLEUKNZGZTFWWWFLGTSLLWJWAMFUQ

Index of Coincidence: 0.0679

g Value	Mg for y1	Mg for y2	Mg for y3	Mg for y4	Mg for y5	Mg for y6	Mg for y7
0	0.031	0.033	0.034	0.038	0.062	0.036	0.037
1	0.034	0.035	0.034	0.035	0.036	0.041	0.033
2	0.033	0.031	0.036	0.040	0.029	0.033	0.033
3	0.028	0.042	0.035	0.033	0.031	0.044	0.033
4	0.038	0.031	0.030	0.028	0.043	0.036	0.047
5	0.031	0.028	0.039	0.034	0.032	0.036	0.027
6	0.031	0.038	0.028	0.034	0.034	0.029	0.029
7	0.034	0.063	0.030	0.043	0.034	0.032	0.039
8	0.061	0.038	0.037	0.037	0.032	0.034	0.061
9	0.039	0.026	0.062	0.041	0.032	0.033	0.034
10	0.033	0.030	0.035	0.031	0.035	0.040	0.028

11	0.032	0.040	0.029	0.039	0.039	0.029	0.030
12	0.036	0.036	0.031	0.035	0.036	0.032	0.038
13	0.032	0.035	0.041	0.037	0.040	0.035	0.034
14	0.029	0.036	0.032	0.032	0.036	0.059	0.038
15	0.038	0.030	0.032	0.035	0.045	0.039	0.033
16	0.029	0.032	0.036	0.031	0.037	0.035	0.028
17	0.032	0.036	0.032	0.032	0.032	0.029	0.033
18	0.035	0.037	0.031	0.043	0.030	0.037	0.037
19	0.044	0.040	0.034	0.033	0.037	0.032	0.043
20	0.036	0.039	0.039	0.029	0.036	0.034	0.038
21	0.038	0.035	0.036	0.032	0.027	0.034	0.037
22	0.037	0.041	0.039	0.058	0.038	0.030	0.033
23	0.046	0.038	0.035	0.039	0.028	0.034	0.040
24	0.036	0.030	0.043	0.032	0.032	0.033	0.035
25	0.033	0.029	0.036	0.027	0.036	0.042	0.029

Decrypting Process

This section details the process used to decrypt the Vigenère cipher text and deduce the key using the Chi-squared test. The Chi-squared test is applied to each substring generated by dividing the cipher text according to the assumed key length. For each substring, the test is used to compare the observed frequencies of letters (after applying a shift) to the expected frequencies of letters in standard English text. The shift that results in the closest match to the expected frequencies (i.e., the lowest Chi-squared value) indicates the letter of the key for that particular substring. Repeating this process for all substrings yields the complete key.

Deduced Key: The process led to the deduction of the key 'STREAMS'. This key is used to decrypt the cipher text, suggesting it was encrypted with a Vigenère cipher using this key.

Decryption Process

After deducing the key, the next step is to decrypt the cipher text. The Vigenère cipher decryption process involves reversing the encryption steps. For each letter in the cipher text, the corresponding letter in the key is used to determine the original letter in the plaintext. This is done by subtracting the position of the key letter from the position of the cipher text letter, then taking the modulo with the alphabet length to handle wrap-around. The result is the original plaintext letter. This process is repeated for each letter in the cipher text to reveal the decrypted text.

Decrypted Text:

THEDEPARTMENTOFJUSTICEHASBEENANDWILLALWAYSBECOMMITTEDTOPROTECTINGT
HELIBERTYANDSECURITYOFTHOSEWHOMWESERVEINRECENTMONTHSHOWEVERWEHA
VEONANEWSCALESEENMAINSTREAMPRODUCTSANDSERVICESDESIGNEDINAWAYTHATG
IVESUSERSSOLECONTROLOVERACCESSTOHEIRDATAASARESULTLAWENFORCEMENTI
SSOMETIMESUNABLETORECOVERTHECONTENTOFELECTRONICCOMMUNICATIONSFRO
MTHETECHNOLOGYPROVIDEREVENINRESPONSETOACOURTORDERORDULYAUTHORIZE
DWARRANTISSUEDBYAFEDERALJUDGEFOREXAMPLEMANYCOMMUNICATIONSSERVICES
NOWENCRYPTCERTAINCOMMUNICATIONSBYDEFAULTWITHTHEKEYNECESSARYTODECR
YPTTHECOMMUNICATIONSSOLELYINTHEHANDSOFTHEENDUSERTHISAPPLIESBOTHWHE
NTHEDATAISINMOTIONOVERELECTRONICNETWORKSORATRESTONANELECTRONICDEVI
CEIFTHECOMMUNICATIONSPROVIDERISSERVEDWITHAWARRANTSEEKINGTHOSECOMM

UNICATIONS THE PROVIDER CANNOT PROVIDE THE DATA BECAUSE IT HAS DESIGNED THE TECHNOLOGY SUCH THAT IT CANNOT BE ACCESSED BY ANY THIRD PARTY. WE DO NOT HAVE ANY SILVER BULLETS AND THE DISCUSSIONS WITHIN THE EXECUTIVE BRANCH ARE STILL ONGOING. WHILE THERE HAS NOT YET BEEN A DECISION WHETHER TO SEEK LEGISLATION, WE MUST WORK WITH CONGRESS, INDUSTRY, ACADEMICS, PRIVACY GROUPS AND OTHERS TO CRAFT AN APPROACH THAT ADDRESSES ALL OF THE MULTIPLE COMPETING CONCERNS THAT HAVE BEEN THE FOCUS OF SO MUCH DEBATE. BUT WE CAN ALL AGREE THAT WE WILL NEED ONGOING, HONEST, AND INFORMED PUBLIC DEBATE ABOUT HOW BEST TO PROTECT LIBERTY AND SECURITY IN BOTH OUR LAWS AND OUR TECHNOLOGY.

Decrypted Text with punctuation:

THE DEPARTMENT OF JUSTICE HAS BEEN AND WILL ALWAYS BE COMMITTED TO PROTECTING THE LIBERTY AND SECURITY OF THOSE WHOM WE SERVE. IN RECENT MONTHS, HOWEVER, WE HAVE ON A NEW SCALE SEEN MAINSTREAM PRODUCTS AND SERVICES DESIGNED IN A WAY THAT GIVES USERS SOLE CONTROL OVER ACCESS TO THEIR DATA AS A RESULT. LAW ENFORCEMENT IS SOMETIMES UNABLE TO RECOVER THE CONTENT OF ELECTRONIC COMMUNICATIONS FROM THE TECHNOLOGY PROVIDER, EVEN IN RESPONSE TO A COURT ORDER OR DULY AUTHORIZED WARRANT ISSUED BY A FEDERAL JUDGE. FOR EXAMPLE, MANY COMMUNICATIONS SERVICES NOW ENCRYPT CERTAIN COMMUNICATIONS BY DEFAULT WITH THE KEY NECESSARY TO DECRYPT THE COMMUNICATIONS SOLELY IN THE HANDS OF THE END USER. THIS APPLIES BOTH WHEN THE DATA IS IN MOTION OVER ELECTRONIC NETWORKS OR AT REST ON AN ELECTRONIC DEVICE. IF THE COMMUNICATIONS PROVIDER IS SERVED WITH A WARRANT SEEKING THOSE COMMUNICATIONS, THE PROVIDER CANNOT PROVIDE THE DATA BECAUSE IT HAS DESIGNED THE TECHNOLOGY SUCH THAT IT CANNOT BE ACCESSED BY ANY THIRD PARTY. WE DO NOT HAVE ANY SILVER BULLETS AND THE DISCUSSIONS WITHIN THE EXECUTIVE BRANCH ARE STILL ONGOING. WHILE THERE HAS NOT YET BEEN A DECISION WHETHER TO SEEK LEGISLATION, WE MUST WORK WITH CONGRESS, INDUSTRY, ACADEMICS, PRIVACY GROUPS AND OTHERS TO CRAFT AN APPROACH THAT ADDRESSES ALL OF THE MULTIPLE COMPETING CONCERNS THAT HAVE BEEN THE FOCUS OF SO MUCH DEBATE. BUT WE CAN ALL AGREE THAT WE WILL NEED ONGOING, HONEST, AND INFORMED PUBLIC DEBATE ABOUT HOW BEST TO PROTECT LIBERTY AND SECURITY IN BOTH OUR LAWS AND OUR TECHNOLOGY.

The Department of Justice has been, and will always be, committed to protecting the liberty and security of those whom we serve. In recent months, however, we have on a new scale seen mainstream products and services designed in a way that gives users sole control over access to their data. As a result, law enforcement is sometimes unable to recover the content of electronic communications from the technology provider, even in response to a court order or duly authorized warrant issued by a federal judge. For example, many communications services now encrypt certain communications by default, with the key necessary to decrypt the communications solely in the hands of the end user. This applies both when the data is in motion over electronic networks or at rest on an electronic device. If the communications provider is served with a warrant seeking those communications, the provider cannot provide the data because it has designed the technology such that it cannot be accessed by any third party. We do not have any silver bullets, and the discussions within the executive branch are still ongoing. While there has not yet been a decision whether to seek legislation, we must work with Congress, industry, academics, privacy groups, and others to craft an approach that addresses all of the multiple competing concerns that have been the focus of so much debate. But we can all agree that we will need ongoing, honest, and informed public debate about how best to protect liberty and security in both our laws and our technology.